

SatsPath Protocol

SatsPath is an open Bitcoin payment resolver and routing protocol. Wallets can implement it without using a SatsPath-hosted platform.

The protocol flow is:

identifier -> signed public profile -> public payment pointers -> route decision -> QR/payment intent

Public Profile

A `PaymentProfile` contains public information only:

- identifier alias,
- identity public key,
- Lightning Address, LNURL, or BOLT-style public invoice data,
- on-chain public addresses and public descriptor hints,
- Ark server and receiver public key,
- update and expiry timestamps.

The profile is signed by the user's identity key. That identity key does not control funds by itself. Wallets and nodes control funds.

Payment Pointers

`PaymentPointer` is the stable output from resolution/routing. It may contain:

- Lightning Address,
- LNURL-pay callback,
- BOLT11 invoice as data,
- Bitcoin address with public claim policy metadata,
- Ark public pointer.

It must never contain private keys, seed phrases, macaroons, certificates, API secrets, or signing keys.

Mainnet Preview vs Mainnet Execution

Mainnet Preview is supported as a safe public-data mode. A wallet or app can ask SatsPath to resolve a recipient, verify the signed profile, choose a rail, and return a mainnet-compatible payment pointer or QR payload.

Mainnet Preview may return public instructions such as Lightning Address pointers, explicitly requested BOLT11 invoices as data, BIP21 `bitcoin:` URIs, or Ark `satspath:ark` intent URIs. The caller must use its own wallet to pay.

Mainnet Execution is not implemented. SatsPath does not custody funds, does not sign Bitcoin transactions, does not broadcast transactions, and does not store wallet private keys.

Platform

The SatsPath Platform is optional. It can provide:

- email verification for normal users,
- invite links,
- API-based profile publishing,
- public profile lookup.

It must never custody funds and must never see wallet private keys.

Compatibility

BIP-353 discovers payment instructions through DNS. BIP-321/BIP-21 style `bitcoin:` URIs encode payment instructions. SatsPath extends this by adding signed public profiles and route selection across Lightning, on-chain Bitcoin, and Ark.